



SUMMARY

Innovative bioinformatics researcher blending AI, systems biology, and computational modeling to extract biological insight from high-dimensional, multimodal data. Experienced in building explainable deep learning frameworks (CNN, GNN, GAT-ResNet) and applying knowledge graphs, SHAP, and quantum algorithms for biological interpretation. Passionate about designing interpretable, creative, and reproducible pipelines that bridge data-driven discovery with molecular understanding.

EDUCATION

M.Sc. in Bioinformatics

Jamia Millia Islamia | 2025–2027

Coursework: Structural Bioinformatics, Systems Biology, Computational Genomics, AI/ML in Biology

B.Sc. in Bioinformatics

MAKAUT, WB | 2022–2025

Key Projects: Alzheimer's prediction (NACC & ADNI), DNA data storage, Inflammasome biomarkers

Seminar: Protein Fold & Profiling of Protein Sequence (2024)

CGPA: 8.49

TECHNICAL SKILLS

Programming & Data Science: Python, R, MATLAB, C++, cQASM, Bash, basic Perl
Machine Learning & Modelling: Deep learning, SHAP, GNN, SINDy, HMM, clustering, optimisation

Bioinformatics & Computational Biology:

Sequence analysis, structural modelling, DEG analysis, pathway enrichment, biomarkers, DNA data storage

Tools & Frameworks: PyMOL, Cytoscape, MSA tools, TensorFlow, DiagrammeR, ggplot2

Databases & Management: NACC, ADNI, large-scale sequence datasets, reproducible pipelines

Languages: English, Hindi, Urdu, Bengali (fluent); Arabic, French (reading proficiency)

RESEARCH & PROJECT EXPERIENCE

Machine Learning & Bioinformatics in Alzheimer's Disease (2025)

Analyzed NACC datasets (clinical, genetic, neuropathological, MRI, CSF biomarkers). Built GAT-ResNet frameworks integrating multimodal biomarkers and applied explainable AI (SHAP, knowledge graphs) for interpretability.

Algorithmic Approach to DNA Data Storage (2024)

Developed Python algorithms for binary data encoding/decoding into DNA sequences with robust error correction and modular pipeline design.

Variational Quantum Eigensolver Pipeline (2024)

Built hybrid workflows in Python and MATLAB applying optimization methods for computational biology datasets and molecular energy estimation.

Cancer Biomarker Research (2024)

Investigated NLRC4 inflammasome biomarkers; conducted pathway mapping and published findings in Human Gene (Elsevier).

Protein Fold & Profiling of Protein Sequence (2023–2024)

Presented research on protein fold prediction and sequence profiling; designed modular R/Python pipelines and delivered departmental seminar.

BioMoR: An R package for advanced bioinformatics predictive modeling (2025)

Bioinformatics Modeling with Recursion and Autoencoder-Based Ensemble using recursive transformer-inspired architectures, autoencoders, random forests, XGBoost, and stacked ensembles. Includes utilities for cross-validation, calibration, benchmarking, and threshold optimisation.

PUBLICATION

Arshad, M., Mal, C., Das, S., & Bhattacharyya, D. (2024).
The potential of the NLRC4 inflammasome as a cancer biomarker: A pan-cancer investigation. Human Gene, 42(8).
DOI:10.1016/j.humgen.2024.201351

Arshad, M., & Mal, C. (Under review).
MIKU: A multi-modal integrated knowledge-guided unified model for Alzheimer's disease prediction. 3 Biotech.

Arshad, M., & Md, S. A. (Under review).
Deep learning in Huntington's disease: Advances in prediction, biomarkers, and progression modeling. Journal of Huntington's Disease.

Arshad, M. (In submission).
SINDy-MUSE: An interpretable dynamics and structure modeling . IEEE/ACM

LEADERSHIP & ACTIVITIES

Student Leader, NSS Unit (2023–2025):
Led interdisciplinary outreach and community projects.

Auditor, OET (2023–2025):
Conducted quality and compliance checks ensuring data accuracy in large-scale assessments.

RESEARCH & PROJECT EXPERIENCE

Chemical–Biological Dynamics Modeling (2025)

Developed computational models simulating chemical–biological interactions; implemented stochastic and deterministic processes in R.

Monte Carlo Finance (R) (2025)

Created R-based Monte Carlo simulations applicable to stochastic biological systems and quantitative modeling.

SINDy–HMM Framework for Modeling Musical Structure (2025)

Developed interpretable framework combining SINDy and HMM for transient acoustic segmentation; developing SINDy-MUSE (IEEE/ACM submission).

INTERESTS

Interpretable AI, Systems Biology, Knowledge Graphs, Quantum–Biological Modeling, Computational Creativity